

BEC305 Python Programming — Assignment 2 —

Choosing and using the built-in data structures

Module 2: Data Structures: Lists, Dictionaries and Strings

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO2: 13 · L1–L2 0, L3–L4 8, L5–L6 5

Objective

Develop judgement about which container fits a problem, and make the mutable-versus-immutable distinction concrete enough that aliasing, shallow copies and in-place methods stop producing surprises.

Learning outcomes

- Student will be able to choose between list, tuple, dictionary and string on the basis of what the data means.
- Student will be able to predict whether an operation mutates in place or returns a new value.
- Student will be able to diagnose aliasing bugs and apply the correct depth of copy.
- Student will be able to model a real-world object as a nested structure and traverse it.

Part A — Problems (17 marks)

P1. Build a list of at least eight items and demonstrate forward indexing, negative indexing, three different slices, and the partition identity `spam[:n] + spam[n:] == spam`. Explain in one sentence why the identity holds. [C02 | L3 | 2 marks]

P2. Write a program that removes every occurrence of a value from a list by looping forward and deleting. Show that it skips items, then fix it two different ways and verify both against the same input. [C02 | L4 | 2 marks]

P3. Demonstrate every list method covered — index, append, insert, remove, sort — recording the return value of each. Then reproduce the `spam = spam.sort()` bug and explain the resulting `TypeError`. [C02 | L3 | 2 marks]

P4. Sort a list of mixed-case names with a plain `sort()` and with `key=str.lower`. Explain the difference using character codes, and state which ordering a user would expect. [C02 | L4 | 2 marks]

P5. Construct a list of lists and demonstrate the three copying behaviours: plain assignment, a slice copy, and `copy.deepcopy`. Present a table of what each sees after a nested mutation. [C02 | L4 | 2 marks]

- P6.** Attempt to use a list as a dictionary key, record the exact error, then repeat with the equivalent tuple. Explain the role of hashability in one paragraph. [C02 | L4 | 2 marks]
- P7.** Write a character-frequency counter twice: once with an explicit if-not-in-dictionary test and once with.setdefault. Compare line counts and explain what.setdefault replaced. [C02 | L3 | 2 marks]
- P8.** Model a fantasy game inventory as a nested dictionary of at least three players each holding several items. Write functions to add an item, total a player's items, and display the inventory with pprint. [C02 | L5 | 1 marks]
- P9.** Write a text-cleaning routine that strips whitespace, validates with an isX method plus a length guard, and reports which of the two guards rejected each bad input. [C02 | L4 | 1 marks]
- P10.** Demonstrate the join/split round trip on a list containing an element with a space in it, then show a separator choice for which the round trip is not exact and explain why. [C02 | L5 | 1 marks]

Part B — Theory (3 marks)

- T1.** Explain why Python distinguishes mutable from immutable types at all, naming two capabilities immutability enables that mutability cannot. [C02 | L5 | 1 marks]
- T2.** Argue why the mutating list methods return None rather than the list, and what would go wrong if they returned the list instead. [C02 | L5 | 1 marks]
- T3.** Given a problem description of your choice, justify a choice between a list of dictionaries and a dictionary of lists, stating what each makes easy and what each makes awkward. [C02 | L5 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC305_Assignment2.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Correct use of the four containers	6	Every structure chosen with a stated reason that matches the data; all operations correct and outputs verified.	Correct choices with brief justification and minor output errors.	Structures used correctly but chosen by habit rather than reason.	A list used where lookup is by label, or magic index numbers left uncommented.
Mutation, aliasing and copying	6	Every aliasing case demonstrated and explained; correct copy depth chosen and justified; return values of mutating methods recorded.	Aliasing understood with one gap in the copy-depth reasoning.	Demonstrations present but explanations thin.	Slice copy assumed safe for nested data, or mutating method results assigned back.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
String handling and validation	4	Validation guards both length and content, join/split used correctly, the character-set behaviour of strip demonstrated.	Validation correct with minor gaps.	Works on clean input only.	join called on the list, or validation that cannot distinguish empty from invalid.
Modelling and presentation	4	Nested model matches the real object, traversal is clean, output is readable via pprint, and every design decision is justified in one line.	Sound model with adequate explanation.	Model works but its shape does not match the described object.	Flat structure with commented index numbers; unreadable output.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO2	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.